# Report II - SF2930 Regression Analysis

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## Project 2 - Travel insurance

## 1 Introduction and Project Goals

This project explores the modeling of insurance coverage for employees traveling for business purposes, with a particular focus on the potential compensation issued in the event of trip cancellations. The model explored here is built upon claim frequency and severity. The overall risk level for a group is then obtained by combining the frequency and severity estimates, which together determine the premium in the pricing model. As given in the assignment:

$$\textbf{Price} = \gamma_0 \times \prod \gamma_{k,i} \tag{1}$$

Where $\gamma_0$ is the base premium and $\gamma_{k,1}$ $k = 1, ..., M$ are the risk factors corresponding to variable number k and variable group number i. Where each risk factor is estimated as described above.

## 2 Analyses and Model Development

### 2.1 Grouping of Variables

As a first step in building the model, the variables were organized into appropriate groups. This grouping was based on our subjective assessment of what seemed reasonable, combined with an iterative process of trial and error. The objective was to ensure that the groups were internally risk homogeneous, while also guaranteeing that, for each variable, every group contained at least one claim with some cost. For each iteration applied to a given variable, the aim was to reduce the $p$-value as much as possible, while at the same time creating realistic groups, when comparing the significance of a subgroup to the reference subgroup. This approach was intended to construct groups that were as statistically distinct and reasonable as possible. In Table 1 all parameters for all variables are shown.

Table 1: GLM Coefficients for All Variables.

| | | Frequency (Poisson) | | | | Severity (Gamma) | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Var.** | **Group** | **Coef** | **SE** | **z** | **p** | **Coef** | **SE** | **z** | **p** |
| 1 | ≤ 5 (ref.) | — | — | — | — | — | — | — | — |
| | 5−20 | 0.5418 | 0.149 | 3.645 | 0.000 | 0.3907 | 0.377 | 1.036 | 0.300 |
| | 20−100 | 1.4195 | 0.140 | 10.108 | 0.000 | 0.9355 | 0.375 | 2.494 | 0.013 |
| | > 100 | 3.0109 | 0.127 | 23.665 | 0.000 | 0.6860 | 0.344 | 1.993 | 0.046 |
| 2 | Mid (ref.) | — | — | — | — | — | — | — | — |
| | Low | −1.1504 | 0.152 | −7.562 | 0.000 | −0.6399 | 0.252 | −2.543 | 0.011 |
| 3 | ≤ 40 (ref.) | — | — | — | — | — | — | — | — |
| | > 40 | 0.8198 | 0.102 | 8.005 | 0.000 | −1.0452 | 0.264 | −3.953 | 0.000 |
| 4 | Excl. (ref.) | — | — | — | — | — | — | — | — |
| | Not Excl. | 0.9608 | 0.199 | 4.823 | 0.000 | 0.0289 | 0.387 | 0.075 | 0.940 |
| 5 | HIGH (ref.) | — | — | — | — | — | — | — | — |
| | LOW | −0.5545 | 0.136 | −4.067 | 0.000 | −0.3565 | 0.263 | −1.357 | 0.175 |
| 6 | World (ref.) | — | — | — | — | — | — | — | — |
| | Nordics | −1.3583 | 0.581 | −2.338 | 0.019 | −0.0569 | 0.474 | −0.120 | 0.904 |
| | Sweden | −1.4929 | 0.415 | −3.595 | 0.000 | −0.1244 | 0.573 | −0.217 | 0.828 |

1: No. Persons, 2: Activity Code, 3: Company Age, 4: Dangerous Areas, 5: Fin. Rating, 6: Travel Area

## 2.2 Choice of variables

In this project, the selection of variables to include in the model began with an initial assessment of which variables appeared intuitively reasonable in an insurance context. During this stage, the approach was deliberately inclusive, resulting in a relatively large set of candidate variables. Subsequently, a *backward elimination* procedure was applied, where one variable was removed at a time. After each removal, the resulting model was evaluated using the likelihood ratio test, Gini score, as well as the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC), to determine whether the variable should be excluded or not.

The full model form which is the baseline for this process is seen in Table 2, with its summary of statistics given in Table 3.

Table 2: Model variables.

| **Variable** |
|---|
| Activity |
| # Persons |
| Company age |
| Financial Rating |
| Travel Area |
| Dangerous Areas |

Table 3: Statistic summary for the full model.

| **Metric** | **Frequency** | **Severity** |
|---|---|---|
| Log-Likelihood | -146.51 | -401.05 |
| AIC | 313.03 | 822.10 |
| BIC | 339.47 | 837.65 |
| Deviance | 170.08 | 12.06 |
| Pseudo $R^2$ | 1.00 | 0.53 |

Table 4: Variable selection using LRT, AIC/BIC and Gini score.

| | **LRT** | | **ΔAIC** | | **ΔBIC** | | |
|---|---|---|---|---|---|---|---|
| **Variable Removed** | **Freq** | **Sev** | **Freq** | **Sev** | **Freq** | **Sev** | **ΔGini** |
| Activity | 72.02 | 7.86 | 70.02 | 5.86 | 67.37 | 4.31 | -0.032 |
| # Persons | 570.86 | 6.05 | 564.86 | 0.05 | 556.93 | -4.61 | -0.277 |
| Company Age | 60.09 | 16.81 | 58.09 | 14.81 | 55.45 | 13.25 | -0.002 |
| Financial Rating | 18.35 | 1.28 | 16.35 | -0.72 | 13.71 | -2.27 | -0.009 |
| Travel Area | 28.79 | -0.70 | 24.79 | -4.70 | 19.50 | -7.81 | -0.015 |
| Dangerous Areas | 18.11 | -0.45 | 16.11 | -2.45 | 13.47 | -4.01 | 0.000 |

From the results in Table 4, 2 variables are significant on the 5% level for both the Frequency and Severity model according to the Likelihood Ratio Test; **Activity Code** and **Company Age**. Table 4 shows that removing these variables results in a higher AIC and BIC for both the Frequency and Severity model, indicating a worse model. Removing these variables gives lower Gini score indicating that we lose discriminatory power.

**Number of Persons** is the strongest variable for discriminatory power, although insignificant on the 5% level for the severity model. The AIC highly suggests that the variable is kept, while BIC would decrease slightly for the Severity model if removed. Although since BIC penalizes large models, the increase if removed in AIC for both parts of the model and BIC for the Frequency model indicate that the variable should be kept.

**Financial Rating** and **Travelling Area** have a debatable effect on the model since the variables are insignificant for the Severity model and decrease the BIC if removed, although will decrease the Gini score slightly as well and increase AIC and BIC for the Frequency model if removed. Intuitively, **Travelling Area** should play a role in the model risk differentiation. **Financial Rating** is most likely correlated with the stronger variables **Activity Code**, **Company Age** and **Number of Persons** meaning that there is a risk of overfitting with this variable.

**Dangerous Areas** have insignificant Severity model and increase AIC and BIC for the Frequency model, although slightly decrease AIC and BIC for the Severity model if removed. Since there is no improvement in Gini score it suggests that the variable only adds complexity to the model and might result in overfitting.

# 3 Results

From the results in Table 4, the conclusion, through motivation in Section 2.2, is that we want to include the variables; **Activity Code**, **Company Age**, **Number of Persons** and **Travelling Area**. The Gini score for this model can be seen in Figure 1 compared to a model which prices only based on **Number of Persons**. We can see that the final model has better discriminatory power than the other model, and is also less overfitted.

In Table 5 we can see that compared to the larger model in Table 3, all statistics have improved. We can for instance see that the AIC and BIC are almost halved in

value, indicating that the model is less overfitted, which is also to be expected when one removes variables. We can further see that the deviance is substantially smaller, indicating that we have less unexplained variation for the smaller model. Lastly, we can also note that the fairly low pseudo $R^2$ of 0.53 for the full model's severity model, has now increased to a better value of 0.75.

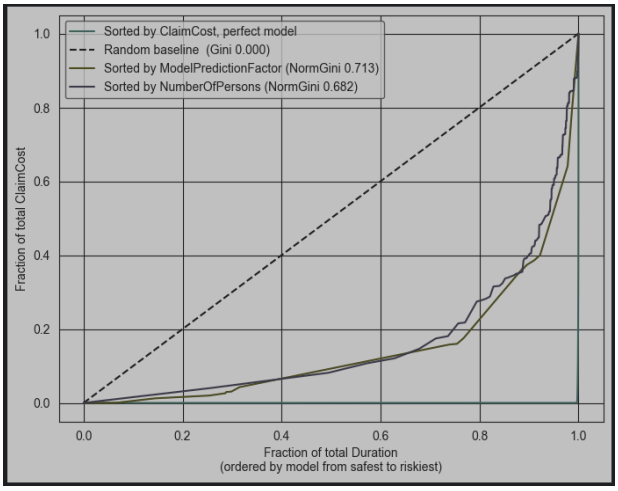



Figure 1: Gini plot for final model compared to model with only Number Of Persons

| **Metric** | **Frequency** | **Severity** |
|---|---|---|
| Log-Likelihood | -74.52 | -224.25 |
| AIC | 163.04 | 462.52 |
| BIC | 172.86 | 469.13 |
| Deviance | 72.00 | 3.70 |
| Pseudo $R^2$ | 1.00 | 0.75 |

Table 5: Statistic summary for final model

### 3.1 Leveling

In Table 6 we can see that year 2020 had higher claims than predicted premium. This is due to the Covid-pandemic during that year, and one can also observe that the claims were lower the year after. Taking this into consideration and looking at Figure 2, the mean risk ratio per year seems to lie around 0.70, meaning that our *target risk ratio* should be somewhere in the range of 0.70. Since one also has to take competition into account, a target risk ratio of 0.75 seems reasonable since this should cover the forecasted claim cost while not yielding a price too high.

With a *base level* of 43.99, the *leveling factor* for the final model is then 58.66. The premium summary for the leveled final model can be seen in Figure 3 and its histogram can be seen in Figure 4.

We can see that most of the customers are companies buying lower-to-moderate premium travel insurance, corresponding to the low-to-moderate-risk groups; for example smaller and newer companies in the Nordics that are in the lower activity risk group. Although worth to note is that we can see a right-tailed distribution where we have some larger companies that are higher risk and therefore pays a higher premium.

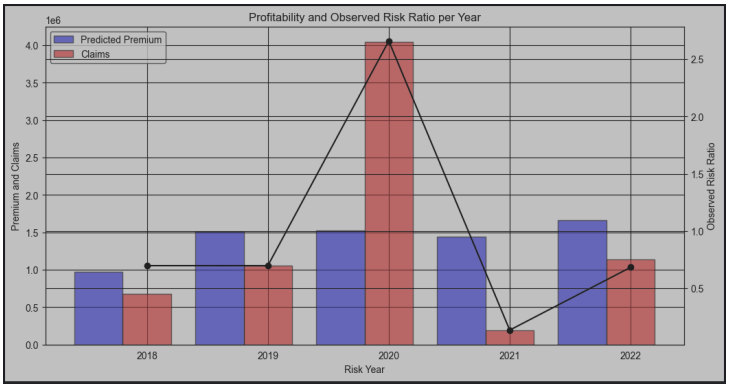



Figure 2: Profitability and Risk Ratio plot.

| Risk Year | Risk Ratio |
|---|---|
| 2018 | 0.70 |
| 2019 | 0.70 |
| 2020 | 2.66 |
| 2021 | 0.13 |
| 2022 | 0.68 |

Table 6: Risk Ratio per Risk Year.

| | |
|---|---|
| count | 79631.000000 |
| mean | 6268.120796 |
| std | 14894.848580 |
| min | 45.924391 |
| 25% | 543.468231 |
| 50% | 2846.162559 |
| 75% | 3573.670116 |
| max | 101677.676870 |

Figure 3: Premium summary for leveled final model

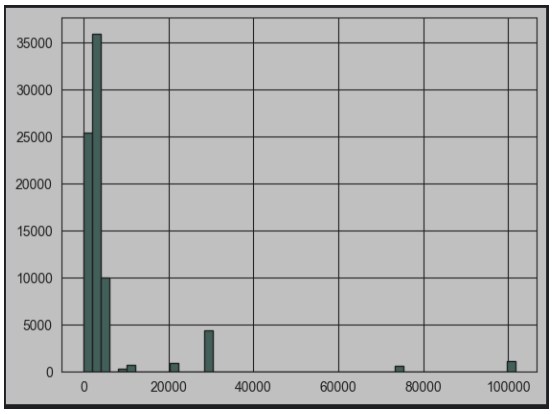


Figure 4: Premium histogram for leveled final model (50 bins).

## Conclusion

The final model was found to include four variables: **Activity Code**, **Company Age**, **Number of Persons**, and **Travel Area**. These variables were selected based on their statistical significance, contribution to discriminatory power (Gini score), and model fit criteria (AIC, BIC, LRT).

The final model demonstrates improved fit compared to the full model, with lower AIC, BIC, and deviance values, as well as an improved pseudo $R^2$ of 0.75 for the severity component. The leveling analysis revealed a target risk ratio of approximately 0.75, accounting for the claims spike in 2020 due to the Covid 19 pandemic and competition. The resulting premium distribution is right skewed, reflecting that the majority of policyholders belong to low risk groups, while a smaller subset of larger, high risk companies require significantly higher premiums.